

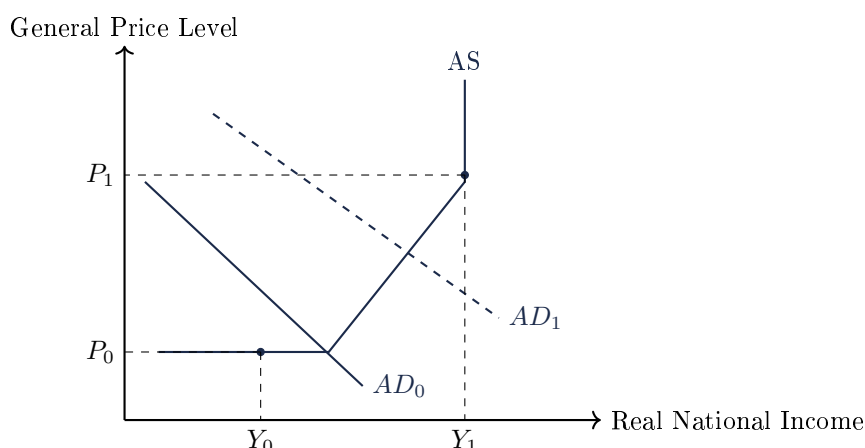
Document 15. Economic Growth — Causes & Consequences

Topic overview. Economic growth is the increase in real GDP (actual growth) or productive capacity (potential growth). Sustained growth requires both to rise in tandem. Sustainable growth additionally requires no environmental depletion; inclusive growth requires fair distribution.

Section A. Important Definitions

Actual Economic Growth	Increase in real national output or real GDP.
Potential Economic Growth	Percentage annual increase in productive capacity of the economy.
Productive capacity	Maximum output the economy can produce with the existing state of technology and available resources.
Sustained Economic Growth	Achieved when both actual and potential growth increase in tandem, leading to a non-inflationary rate of growth.
Sustainable Economic Growth	A rate of economic growth that can be maintained without creating other significant economic problems (depleted natural resources, environmental problems), particularly for future generations.
Inclusive Economic Growth	A rate of economic growth that is sustained over time, is distributed fairly across society, and creates opportunities for the majority of the population.

Section A1. Diagram — Actual Growth via AD



Section B. Framework: Actual Growth via AD (verbatim Crucible)

Step 1. Select the relevant $C/I/G/(X - M)$ factor.

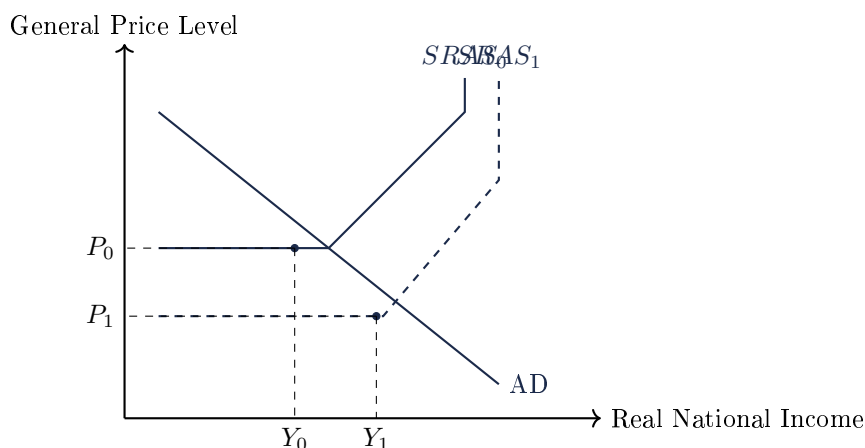
Step 2. Consequently, $C/I/G/(X - M)$ increases / decreases. As $C/I/G/(X - M)$ is a direct component of AD, this causes an / a increase / decrease in AD, ceteris paribus, represented by a rightward / leftward shift in the AD curve.

Step 3. The increase / decrease in AD creates shortages / surpluses in the economy, and firms experience an unplanned fall / rise in inventories. Producers will thus increase / decrease their production level in the next period to maintain inventories at a desired level.

Step 4. To increase / decrease production levels, more / less workers are hired because the demand for labour is a derived demand.

Step 5. This leads to a more than proportionate increase / decrease in real national income from Y_0 to Y_1 via the multiplier effect, resulting in positive / negative actual economic growth.

Section C0. Diagram — Actual Growth via SRAS Increase



Section C. Framework: Actual Growth via SRAS (verbatim Crucible)

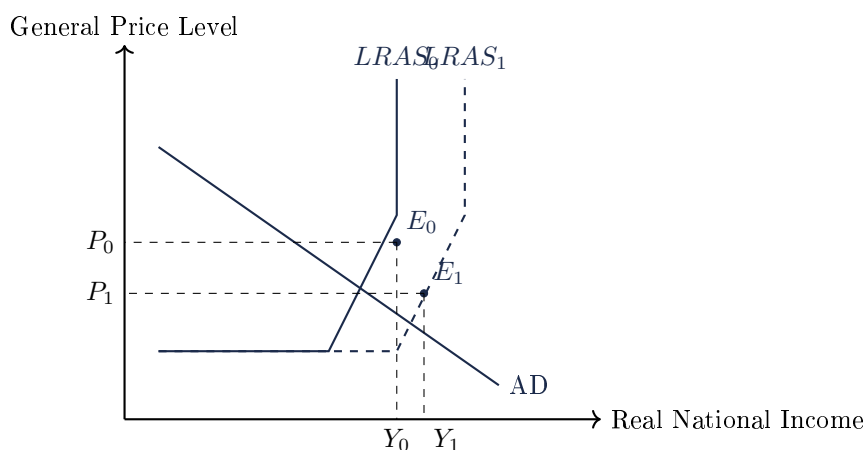
Step 1. Select the relevant SRAS factor and explain how it increases costs of production.

Step 2. This leads to an / a increase / decrease in SRAS, represented by a downward / upward shift in the SRAS curve from $SRAS_0$ to $SRAS_1$. Consequently, surpluses / shortages result in the economy which causes general price levels to decrease / increase from P_0 to P_1 .

Step 3. As firms find it more / less profitable to produce, an / a increase / decrease in output results in the economy, represented by the change in real national income from Y_0 to Y_1 .

Step 4. Consequently, actual economic growth increases / decreases.

Section D0. Diagram — Actual Growth via LRAS Increase



Section D. Framework: Actual Growth via LRAS (verbatim Crucible)

Step 1. Select the relevant LRAS factor.

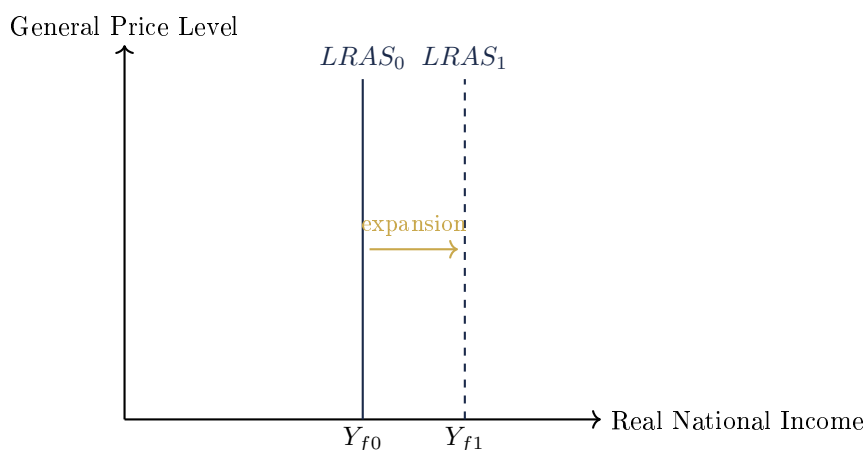
Step 2. This leads to increased / decreased productive capacity represented by the rightward / leftward shift in the LRAS curve from $LRAS_0$ to $LRAS_1$.

Step 3. The increase in potential growth contributes to a faster rate of actual growth as the economy

was initially producing on the intermediate / classical range where resources were increasingly scarce / fully employed. Hence, further increases in AD led to dampened / no increases in actual growth.

Step 4. As the economy comes to a new equilibrium at E_1 , real national income increases from Y_0 to Y_1 , resulting in actual economic growth.

Section E0. Diagram — Potential Growth (LRAS Rightward Shift)



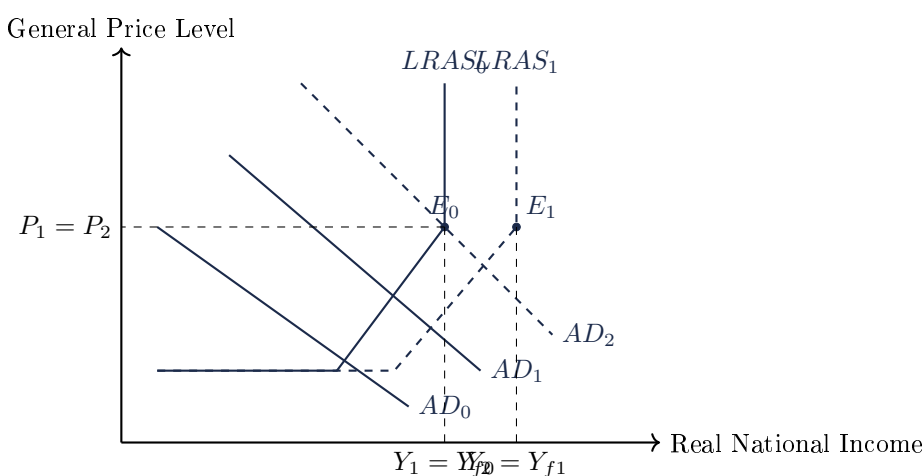
Section E. Framework: Potential Growth (verbatim Crucible)

Step 1. Select the relevant LRAS factor.

Step 2. This leads to increased / decreased productive capacity represented by the rightward / leftward shift in the LRAS curve from $LRAS_0$ to $LRAS_1$.

Step 3. Thus, potential economic growth increases / decreases from Y_{f0} to Y_{f1} .

Section F. Diagram — Sustained Economic Growth (Actual + Potential in tandem)



Section G. Framework: Sustained Economic Growth (verbatim Crucible)

Step 1. Initially, real national output is at Y_0 and the economy is experiencing slow or even negative economic growth.

Step 2. An increase in AD from AD_0 to AD_1 leads to an increase in real national output from Y_0 to Y_1 , and actual growth is achieved.

Step 3. At Y_1 , if AD continues to increase without increases in LRAS, it will result in demand-pull inflation from P_1 to P_3 with no corresponding increase in real national output.

Step 4. For sustained economic growth to occur, there must be an increase in productive capacity and LRAS must increase from $LRAS_0$ to $LRAS_1$. As LRAS increases together with AD, both actual and potential economic growth increase. Actual economic growth increases from Y_1 to Y_2 , and potential economic growth increases from Y_{f0} to Y_{f1} .

Step 5. Furthermore, general price levels remain unchanged as $P_1 = P_2$, resulting in non-inflationary growth.

Step 6. Thus, sustained economic growth is achieved.

Section H. Consequences of Sustained Economic Growth

Benefits:

- Higher tax revenue → budget surplus → allows fiscal + supply-side policies to reinforce future growth, inclusive growth (Workfare, transfers), sustainable growth (Resource Efficiency Grant for Energy — REG(E)).
- Lower inflation pressure as LRAS expansion eases factor scarcity.
- Lower cyclical unemployment as AD rises; lower structural unemployment as budget allows retraining (SkillsFuture).
- BOP: improved current account via lower-cost / higher-quality exports; improved capital + financial account via stronger investment climate (FDI).
- Higher material and non-material SOL.

Costs:

- Income inequality may worsen — rich gain disproportionately from new business opportunities; inclusive growth not guaranteed.
- Environmental damage (pollution, resource depletion) worsens sustainable growth (e.g. China rapid industrialisation).
- Demand-pull inflation if AD outpaces LRAS in intermediate/classical range.
- Structural unemployment may rise as economy restructures (Singapore 1980s electronics).
- Current account may worsen — FDI repatriation, higher import demand ($YED > 0$).
- Non-material SOL may worsen (stress, pollution, longer working hours).

Section I. Singapore Application

- **Sustained:** SkillsFuture, RIE 2025 (S\$25bn), EDB FDI promotion, BTO/MRT expansion.
- **Sustainable:** Green Plan 2030, carbon tax (S\$25 → S\$45 by 2026–27), Resource Efficiency Grant for Energy.
- **Inclusive:** Progressive Wage Model, Workfare Income Supplement, ComCare, GST Vouchers + Assurance Package.

NOTE: Frameworks reproduced word-for-word from the Crucible Education Centre handout.